

PIXIU TECH LLP

Robotics Lab Book

Class 7 | Sensing the Environment

Control light with a knob, measure the air in your classroom, and build an alarm that knows how hard it is raining.

FIVE UNITS Introduction | Arduino IDE | Basic | Intermediate | Final

FIFTEEN SESSIONS One session = one class

THREE PROJECTS LED Dimmer - Temperature and Humidity Monitor - Rain Alarm

BOARD Arduino Uno

THIS BOOK BELONGS TO

SCHOOL / SECTION

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Fifteen sessions. One session is one class period.

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When you see	It means
Today you will	The three things this class is about. Read it before you start.
Try This	A short activity to do right now, at your desk, with your kit.
Watch Out	A mistake almost everyone makes. Read it before you wire.
Did You Know	A fact worth taking home and telling someone.
Dark box	Arduino code. Type it exactly - every bracket and semicolon matters.
Challenge	For when you finish early and want something harder.

UNIT 1

Introduction

Sessions 1 and 2

Last year your machines answered yes or no. This year they will answer *how much*. That single change - from on-or-off to a whole range of values - is what turns a gadget into a real sensing machine.

By the end of this unit you will be able to:

- Explain the difference between a digital signal and an analog signal.
- Read a resistor's value from its colour bands.
- Recognise the VCC, GND, DO and AO pins that every sensor module uses.
- Name every component in your Class 7 kit and say what it is for.

SESSION 1

one class

The Digital World and the Analog World

Today you will

- Understand why some signals are on-or-off and others are a range.
- Learn where each idea is used in real machines.
- Meet the twelve components in your kit.

Think about a light switch on your classroom wall. It has exactly two positions: on and off. There is no halfway. Engineers call that a **digital** signal - it can only ever be one of two values, and there is nothing in between.

Now think about a fan regulator, or the brightness slider on a phone. These do not jump between two states. They move smoothly through hundreds of positions. Engineers call that an **analog** signal - a value anywhere within a range.

Here is why this matters. The real world is almost entirely analog. Temperature does not jump from cold to hot; it climbs. Light does not switch from day to night; it fades. Rain is not simply raining or not raining; there is drizzle, and there is a downpour. If your machine can only understand on-or-off, it can never describe the world properly.

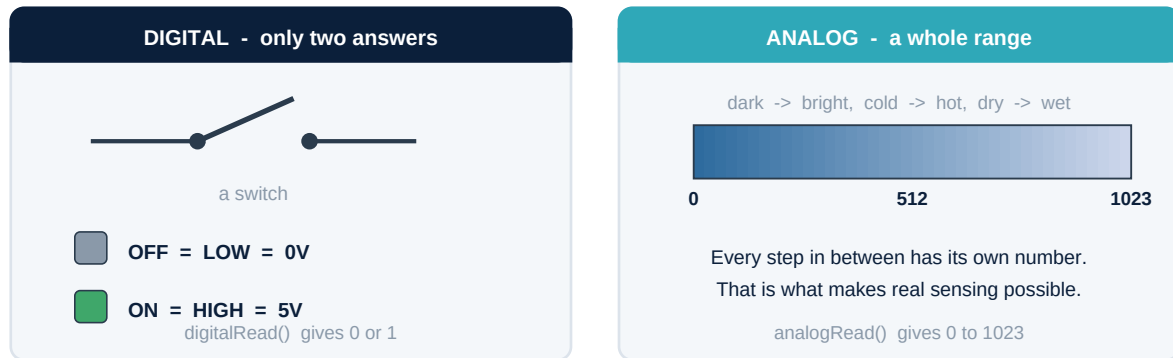


Figure 1.2 - The big idea of Class 7: numbers instead of yes-or-no

The Arduino handles both. Its digital pins, D0 to D13, deal in HIGH and LOW. Its analog pins, A0 to A5, measure a voltage and turn it into a number from 0 to 1023. Every project in this book depends on that second row of pins - which is exactly why Class 7 goes further than Class 6 did.

Try This

In your notebook, make two columns headed DIGITAL and ANALOG. Put these into the right column: a doorbell, a volume knob, a tubelight switch, room temperature, a keyboard key, the fuel gauge in a car, sunlight through a window. Be ready to defend any answer you are unsure about.

1.1 Open the Box

Lay everything out on the desk before you plug anything in. Learn the names first - it is impossible to ask a trainer for help with 'that small blue thing'. Notice that several of these parts are new this year: the potentiometer, the DHT11, the rain sensor and the servo motor.

ARDUINO KIT COMPONENTS – CLASS 7

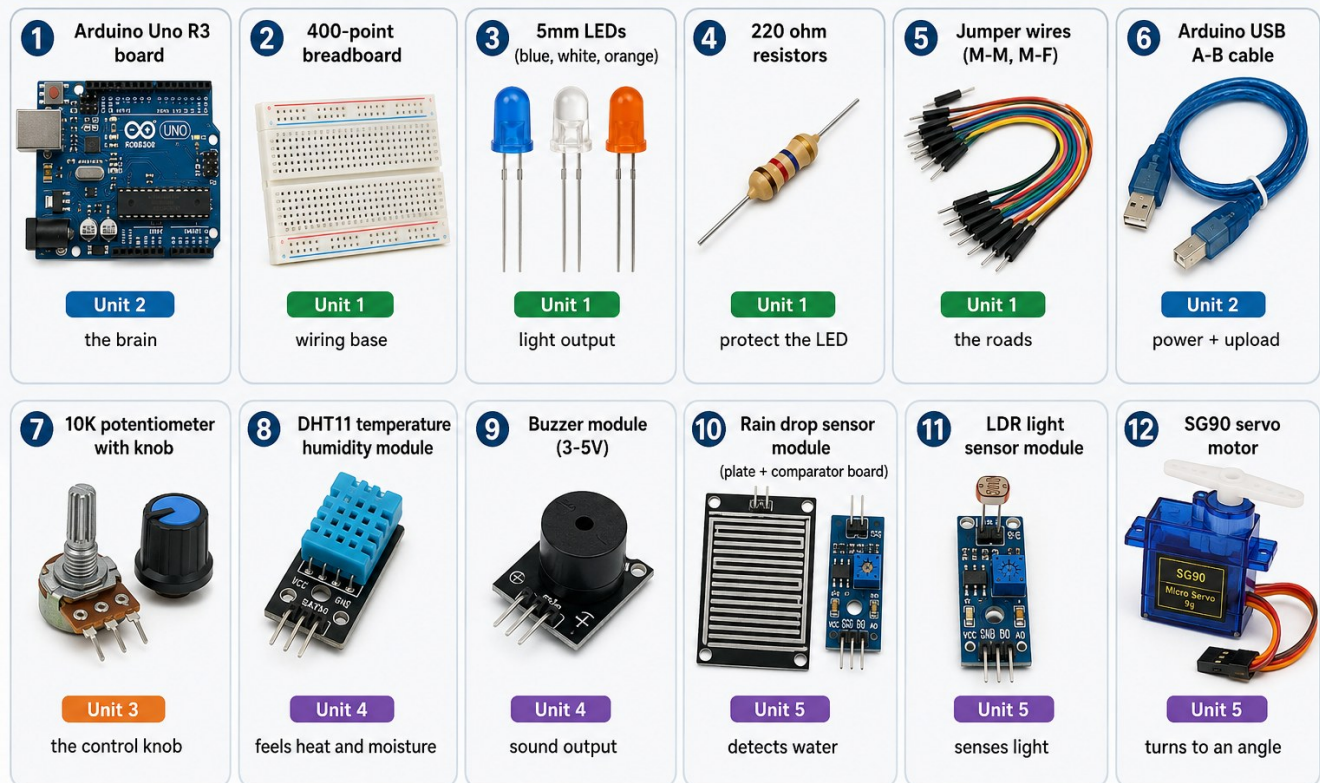


Figure 1.1 - Your Class 7 kit. Learn these twelve names before the next class.

Part	What it does	First used in
Arduino Uno board	The brain. It stores your program and runs it forever, even after the power has been off.	Session 3
USB A-B cable	Carries your program into the board and powers it while you test.	Session 3
400-point breadboard	Joins components together without any soldering.	Session 2
Jumper wires	The roads between parts. Male-to-male and male-to-female.	Session 2
5mm LEDs	Tiny lamps. They glow only one way round.	Session 2
220 ohm resistors	Limit the current so an LED is not destroyed.	Session 2
10K potentiometer	A knob that gives a smoothly changing value - your first analog input.	Session 6
DHT11 module	Measures temperature and humidity, both from one sensor.	Session 9
Buzzer module	Makes sound so the machine can warn people.	Session 11
Rain drop module	A plate that gets wet, plus a small board that reports how wet it is.	Session 12
LDR light sensor module	Reports how bright the surroundings are.	Session 13
SG90 servo motor	Turns to an exact angle and holds it. Used in the challenge work.	Session 15

Watch Out

The rain sensor comes in **two pieces**: the black plate with gold lines, and a small green control board. They join with two wires. Only the plate is allowed to get wet - never the control board, never the Arduino.

New word	What it means
Digital	A signal with only two possible values: on or off.
Analog	A signal that can take any value within a range.
Module	A small ready-made board with the fiddly parts already fitted.

SESSION 2

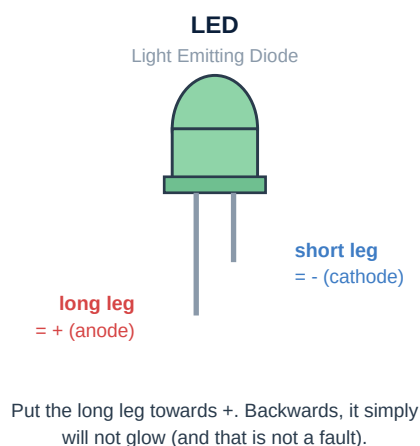
one class

Components, Resistor Colours and Modules

Today you will

- Read a resistor's value from its colour bands.
- Learn the four pins that almost every sensor module uses.
- Rebuild the LED circuit from Class 6 as a warm-up.

A resistor is a small part with a big job: it limits how much current can flow. Its value is measured in ohms, and it is printed on the body as coloured bands rather than numbers, because numbers would be too small to read. Each colour stands for a digit.



Its job is to protect the LED from too much current, like a narrow gate in a wide road.

Colour	Digit	Colour	Digit
Black	0	Green	5
Brown	1	Blue	6
Red	2	Violet	7
Orange	3	Grey	8
Yellow	4	White	9

The first two bands give you two digits. The third band tells you how many zeros to add after them. The fourth band, usually gold, is about accuracy and you can ignore it for now. So a resistor banded **red - red - brown** reads as 2, then 2, then one zero: **220 ohms**. A resistor banded **brown - black - orange** reads 1, then 0, then three zeros: **10,000 ohms**, which everybody calls 10K.

Try This

Find the 220 ohm and the 10K resistors in your kit without looking at the packet. Hold each one up and read the bands aloud to your partner. Then swap. This takes two minutes and saves you from a wrong resistor in every project this year.

2.1 The Four Pins Every Module Has

Look closely at the DHT11, the LDR module and the rain sensor board. They look completely different, but their pins follow the same pattern. Learn this pattern once and you can wire any sensor module you ever meet, including ones that do not exist yet.

Pin	Full name	What it is for	Connect it to
VCC	Voltage at the Common Collector	Power into the module	5V on the Arduino
GND	Ground	The return path for the electricity	GND on the Arduino
DO	Digital Out	A yes-or-no answer, set by the small blue screw on the board	any digital pin, D2 to D13
AO	Analog Out	A full reading from 0 to 1023	an analog pin, A0 to A5

Not every module has all four. The DHT11 has no AO pin at all, because it is a digital sensor that sends its readings as a coded pulse - you will meet that in Unit 4. But VCC and GND are always there, on every module, without exception.

Watch Out

The small blue box with a screw on many modules is a **trimmer**. Turning it changes the level at which the DO pin flips from LOW to HIGH. It has no effect at all on the AO pin. If you are using AO and someone tells you to 'adjust the screw', they are giving you the wrong advice.

Warm-up Activity - The Class 6 Circuit, From Memory

Before the coding starts again, prove your hands still remember. Build this without looking at any diagram.

- 1 Push an LED into the breadboard with its two legs in different columns.
- 2 Put a 220 ohm resistor in the same column as the long leg.
- 3 Wire the free end of the resistor to 5V, and the LED's short leg column to GND.
- 4 Plug in the USB. The LED should glow with no code at all.
- 5 Now move the resistor to the **short** leg side instead. Does it still work? Why?

Did You Know

Step 5 works perfectly. A resistor limits the current in the whole loop, so it does the same job wherever you place it in that loop. What matters is that it is *somewhere* in the path, not which side it sits on.

New word	What it means
Ohm	The unit of resistance.
Trimmer	A tiny adjustable screw that sets a module's DO threshold.
VCC	The power-in pin of a module. Goes to 5V.
AO	The analog-out pin. Gives a full reading, not just yes or no.

CHECK WHAT YOU LEARNT

Check Yourself - Unit 1**A. Choose the correct answer**

1. A signal that can only be ON or OFF is called
(a) analog (b) digital (c) electric (d) linear
2. Room temperature is best described as
(a) digital (b) analog (c) neither (d) both
3. A resistor banded red - red - brown has a value of
(a) 22 ohm (b) 220 ohm (c) 2200 ohm (d) 10K
4. The VCC pin of a sensor module connects to
(a) GND (b) 5V (c) A0 (d) D13
5. The AO pin of a module gives
(a) only 0 or 1 (b) a number from 0 to 1023 (c) 5 volts always (d) nothing
6. The small blue screw on a module adjusts
(a) the AO reading (b) the DO threshold (c) the power (d) the colour

B. Fill in the blanks

- 1 Analog pins on the Arduino Uno are labelled _____ to _____.
- 2 The third colour band on a resistor tells you how many _____ to add.
- 3 Brown - black - orange is a _____ ohm resistor.
- 4 Every sensor module must always have _____ and GND connected.

C. Answer in one or two lines

- 1 Give two examples of analog quantities in your own home, and say why they are analog.
- 2 Why can a machine that understands only digital signals never describe rain properly?
- 3 A friend says the AO reading changed after turning the blue screw. Are they right? Explain.
- 4 Write down what red - red - brown means, digit by digit.

UNIT 2

The Arduino IDE

Sessions 3, 4 and 5

You already know how to make a pin go on and off. This unit teaches the three skills that separate Class 7 from Class 6: reading a number from a sensor, printing it so you can see it, and turning it into a smooth output.

By the end of this unit you will be able to:

- Upload a program and use the Serial Monitor to watch live values.
- Explain what `analogRead()` returns and why the range is 0 to 1023.
- Use `map()` to convert one range of numbers into another.
- Use `analogWrite()` and PWM to fade an output instead of switching it.

SESSION 3

one class

The Board, the Software and the First Upload

Today you will

- Refresh the parts of the Arduino Uno, especially the analog pins.
- Set the board and port correctly in the IDE.
- Upload Blink and confirm your whole setup works.

The Arduino Uno is a small computer with no screen and no keyboard. It runs exactly one program at a time, keeps it in memory when the power is removed, and starts running again the instant power returns. This year, pay particular attention to two groups of pins: the analog inputs along the bottom, and the digital pins marked with a small ~ symbol.

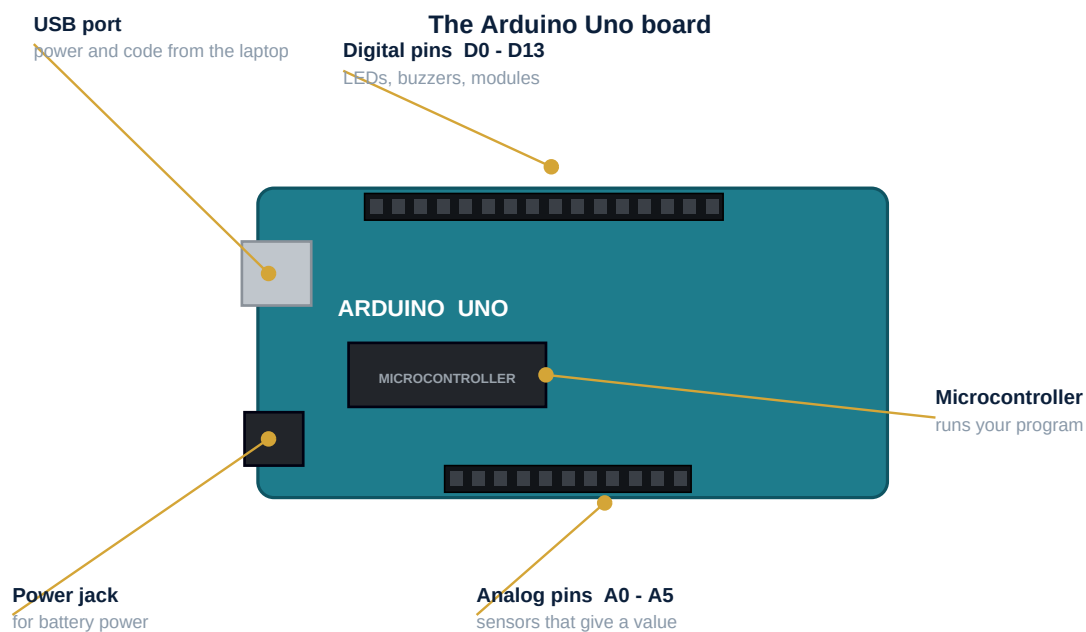


Figure 2.1 - Know your board before you wire anything to it

Group of pins	What is special about them	Used in this book for
A0 to A5	They measure a voltage and report 0 to 1023	the potentiometer, the LDR, the rain sensor
~3, ~5, ~6, ~9, ~10, ~11	They can fade an output, not just switch it	the LED dimmer and mood light
D2 to D13 (all)	Plain on-or-off outputs and inputs	LEDs, the buzzer, the DHT11
5V and GND	Power out to every module you connect	everything, always

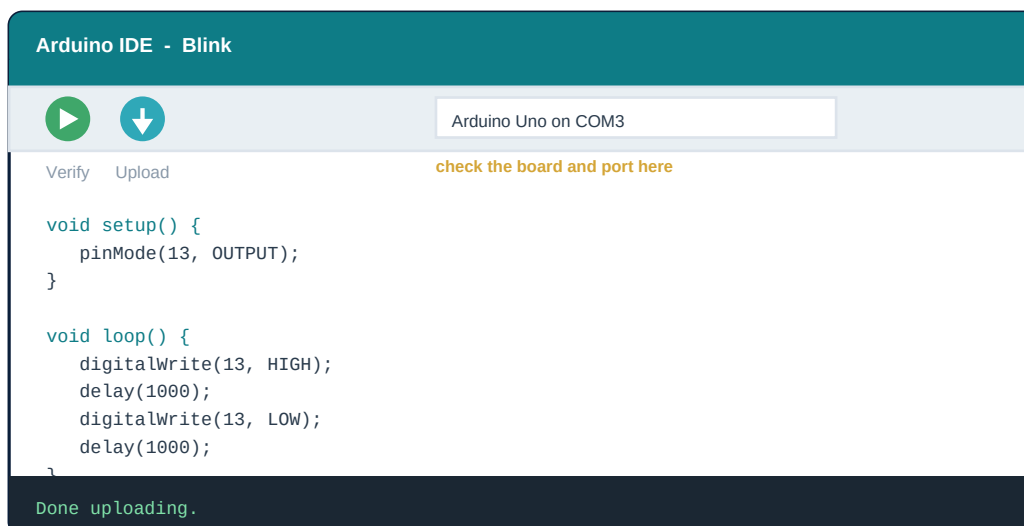


Figure 2.2 - Verify checks your code; Upload sends it to the board

Watch Out

Before your first upload of the year, check **Tools > Board** says Arduino Uno and **Tools > Port** has a port selected. When an upload fails, this is the first thing to check, every single time.

Upload the Blink sketch from Class 6 as a test. If pin 13 blinks, your board, your cable, your laptop and your software are all working. Never start a new project without knowing that.

SESSION 4

one class

Variables and the Serial Monitor**Today you will**

- Learn what a variable is and why int and float are different.
- Open the Serial Monitor and watch live numbers.
- Read your first real sensor value from the potentiometer.

A **variable** is a named box that holds a number. You have already used them: `int led = 13;` makes a box called led and puts 13 in it. This year your boxes will hold values that keep changing, because they come from sensors.

You write	It holds	Example
int	A whole number, with no decimal point	<code>int light = 743;</code>
float	A number with a decimal point	<code>float temp = 27.4;</code>
boolean	Only true or false	<code>boolean raining = false;</code>

Choose the right type. A temperature of 27.4 degrees stored in an `int` becomes 27, and the .4 is lost forever. That does not matter much for a classroom thermometer, but it would matter a great deal in a hospital.

4.1 Talking to the Laptop

Until now your machine has had no way of telling you what it is thinking. The Serial Monitor fixes that. Two instructions are all you need: `Serial.begin(9600);` inside `setup()`, which opens the conversation, and `Serial.println(value);` inside `loop()`, which sends one number and starts a new line.

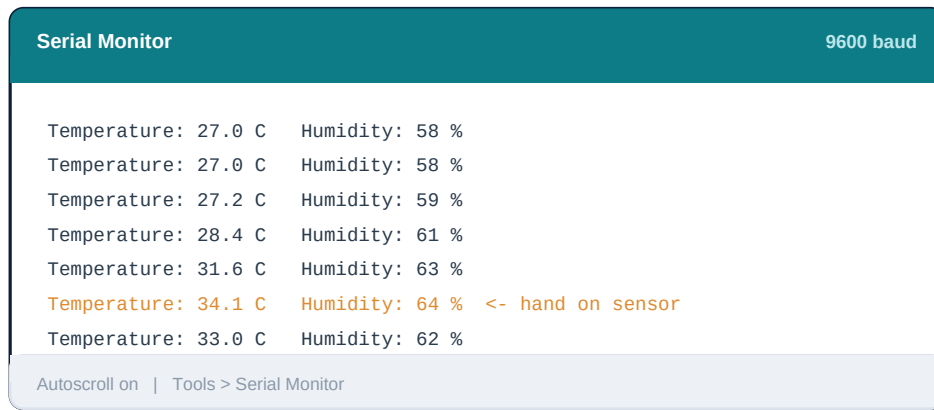


Figure 2.3 - The Serial Monitor is your window into the machine's mind

ReadTheKnob.ino

```
int potPin = A0;

void setup() {
  Serial.begin(9600);      // open the line to the laptop
}

void loop() {
  int value = analogRead(potPin);  // 0 to 1023
  Serial.print("Knob reads: ");
  Serial.println(value);           // println starts a new line
  delay(200);
}
```

Try This

Wire the potentiometer: left leg to 5V, right leg to GND, middle leg to A0. Upload the program above and open **Tools > Serial Monitor**. Turn the knob slowly from one end to the other and watch the numbers. Write down the lowest and the highest number you see.

Watch Out

The number at the bottom right of the Serial Monitor window must say **9600**. If it says anything else you will see meaningless symbols instead of numbers. The number in your code and the number in the window have to match.

SESSION 5

one class

analogRead, map and PWM**Today you will**

- Understand why analogRead gives 0 to 1023 and analogWrite needs 0 to 255.
- Use map() to convert between the two ranges.
- Fade an LED smoothly using PWM.

Here is a puzzle that catches almost everybody. `analogRead()` gives you a number from **0 to 1023**. But `analogWrite()` only accepts a number from **0 to 255**. Feed one into the other and your LED reaches full brightness after a quarter of a turn and then does nothing for the rest.

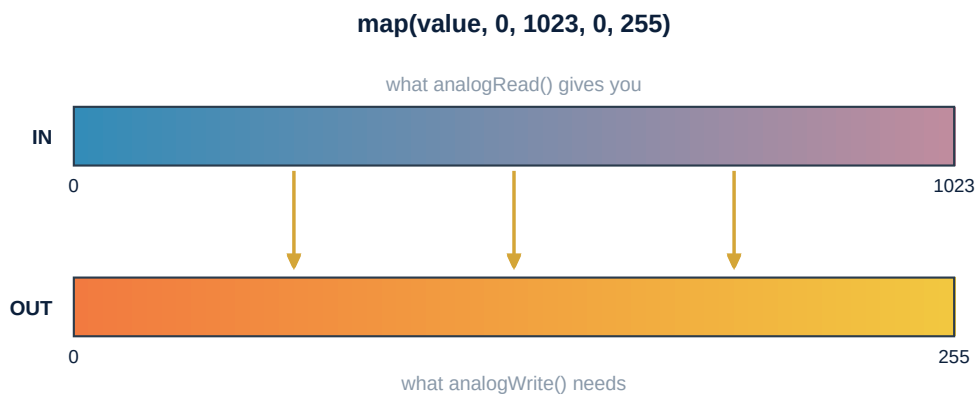


Figure 2.4 - map() stretches one range neatly onto another

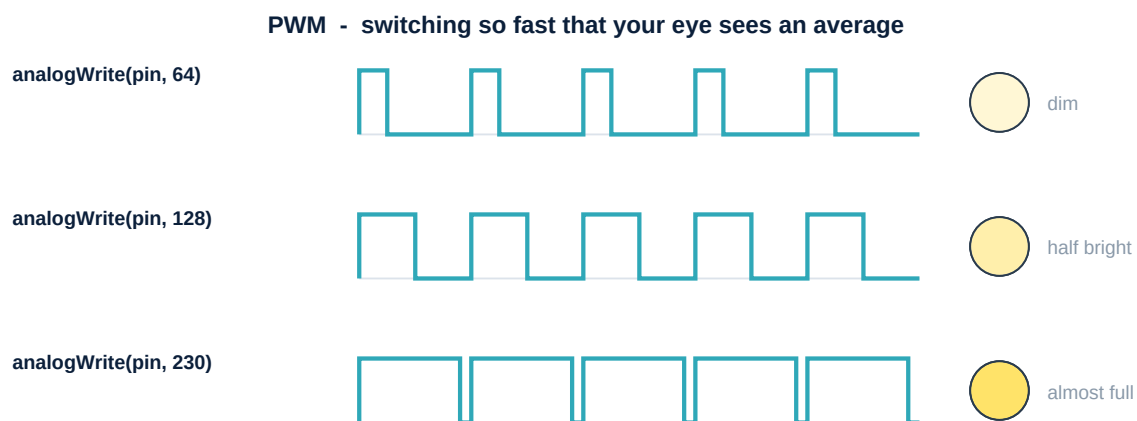
The fix is a single instruction: `map( )`. You tell it the value, the range it came from, and the range you want it in. It does the arithmetic for you.

The three lines that matter

```
int raw = analogRead(A0);           // 0 to 1023
int bright = map(raw, 0, 1023, 0, 255); // now 0 to 255
analogWrite(9, bright);              // pin 9 has a ~ symbol
```

5.1 What PWM Really Does

A digital pin cannot produce half of 5 volts. It can only be fully on or fully off. So how does an LED get dimmer? The board cheats, brilliantly. It switches the pin on and off hundreds of times a second. If the pin is on for a quarter of the time, the LED receives a quarter of the energy, and your eye - which is far too slow to see the switching - simply reports 'dim'.



The pin is still only ON or OFF. It is the amount of time spent ON that changes.

Only pins marked with ~ can do this: D3, D5, D6, D9, D10 and D11.

This trick is called **Pulse Width Modulation**, or PWM, and it is everywhere: in the brightness of your phone screen, in the speed of a drone motor, in the dimmer of an LED bulb. Only six pins on the Uno can do it, and they are all marked with a small ~ next to the number.

Did You Know

PWM is why a cheap LED bulb sometimes looks like it is flickering in a video, even though it looks steady to your eye. The camera's shutter catches the off moments; your eye blurs them together.

New word	What it means
Variable	A named box that holds a number.
int / float	A whole number / a number with a decimal point.
Serial Monitor	The window where a machine prints what it is thinking.
map()	An instruction that stretches one range of numbers onto another.
PWM	Switching a pin on and off very fast so the output appears in-between.

CHECK WHAT YOU LEARNT

Check Yourself - Unit 2**A. Choose the correct answer**

1. `analogRead()` returns a number between
(a) 0 and 255 (b) 0 and 1023 (c) 0 and 5 (d) 0 and 100
2. `analogWrite()` accepts a number between
(a) 0 and 1023 (b) 0 and 255 (c) 0 and 10 (d) 0 and 5
3. `map(raw, 0, 1023, 0, 255)` is used to
(a) read a sensor (b) convert one range to another (c) print a value (d) add a delay
4. PWM works by
(a) lowering the voltage (b) switching on and off very fast (c) using a resistor (d) using the analog pins
5. Which pin can fade an output?
(a) D2 (b) D4 (c) D9 (d) D7
6. `Serial.begin(9600)` belongs inside
(a) `loop()` (b) `setup()` (c) `map()` (d) a comment

B. Fill in the blanks

- 1 A whole number is stored in a variable of type _____.
- 2 A number with a decimal point is stored in a _____.
- 3 PWM stands for Pulse _____ Modulation.
- 4 PWM-capable pins are marked with the symbol _____.

C. Answer in one or two lines

- 1 Explain why 1023 must be converted to 255 before `analogWrite()`.
- 2 Describe in your own words how a pin that is only ever on or off can dim an LED.
- 3 Your Serial Monitor shows strange symbols instead of numbers. What is wrong?
- 4 Why would storing a temperature of 27.4 in an `int` be a bad idea?

UNIT 3

Basic Project: LED Dimmer and Mood Light

Sessions 6, 7 and 8

Your first project of the year has no sensor at all - the sensor is you. A knob becomes an analog input, and three LEDs become an output that changes smoothly rather than jumping. Everything in Units 4 and 5 is built on this.

By the end of this unit you will be able to:

- Wire a potentiometer as a voltage divider and read it.
- Convert the reading into a brightness with `map()`.
- Control an LED smoothly using `analogWrite()`.
- Divide a range into zones and light a different lamp in each.

SESSION 6

one class

Reading the Potentiometer

Today you will

- Learn how a potentiometer turns rotation into voltage.
- Wire it correctly and read it on the Serial Monitor.
- Record the readings at both ends and in the middle.

A potentiometer - everyone calls it a **pot** - is a resistor with a slider. It has three legs. The two outer legs go to 5V and GND, so a fixed voltage sits across the whole strip. The middle leg, called the **wiper**, touches the strip at whatever point the knob has turned to. Turn the knob and the wiper picks up a different voltage, anywhere from 0V to 5V.

That is the whole trick, and it has a name: a **voltage divider**. It is how a volume knob works, how a joystick works, and how a car's accelerator pedal tells the engine how hard you are pressing.

Knob position	Voltage on the wiper	analogRead gives about
Turned fully anticlockwise	0 V	0
Halfway	2.5 V	512
Turned fully clockwise	5 V	1023

Watch Out

If your reading jumps straight from 0 to 1023 with nothing in between, you have almost certainly put the wiper on an outer leg. The middle leg is always the one that goes to A0.

Try This

Reverse the two outer legs - 5V and GND the other way round. Read the values again. The knob now counts backwards. Nothing is broken; you have simply flipped which end of the strip is which. Put it back before the next session.

SESSION 7

one class

Building the Dimmer

Today you will

- Wire the pot and three LEDs on the breadboard.
- Use map() and analogWrite() to control brightness.
- Get a smooth fade across the full turn of the knob.

LED dimmer and mood light - connections

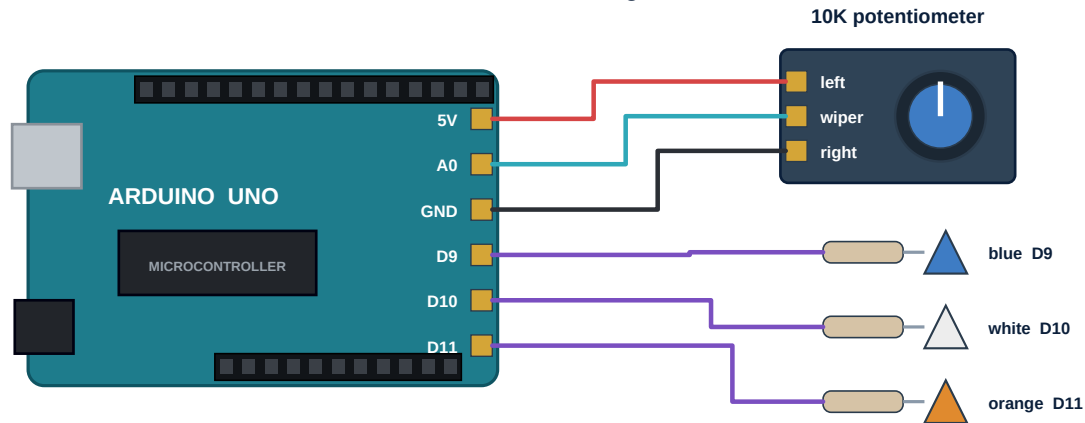


Figure 3.1 - One knob in, three lamps out

Component pin	Arduino pin	Why
Pot - left leg	5V	One end of the strip
Pot - right leg	GND	The other end
Pot - middle leg (wiper)	A0	The changing voltage you read
Blue LED long leg	D9 (~) through 220 ohm	PWM-capable pin
White LED long leg	D10 (~) through 220 ohm	PWM-capable pin
Orange LED long leg	D11 (~) through 220 ohm	PWM-capable pin
All LED short legs	GND	Shared return path

LedDimmer.ino

```

int potPin = A0;
int ledPin = 9;          // must be a pin marked with ~

void setup() {
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int raw = analogRead(potPin);          // 0 to 1023
  int bright = map(raw, 0, 1023, 0, 255); // 0 to 255

  analogWrite(ledPin, bright);           // fade the LED

  Serial.print("raw: ");
  Serial.print(raw);
  Serial.print("  brightness: ");
  Serial.println(bright);

  delay(50);
}

```

Notice `Serial.print` without the `\n` for the first three, and `Serial.println` only on the last one. That keeps all four pieces on a single line and starts a new line at the end. It is a small detail that makes your output readable instead of a mess.

SESSION 8

one class

The Mood Light Challenge

Today you will

- Divide the knob's range into three zones.
- Light a different LED in each zone.
- Test the boundaries and fix the flickering.

A dimmer controls *how much*. A mood light also decides *which*. Split the range from 0 to 1023 into three equal zones and give each zone its own colour. The knob now does two jobs at once: it chooses the lamp and sets the brightness.

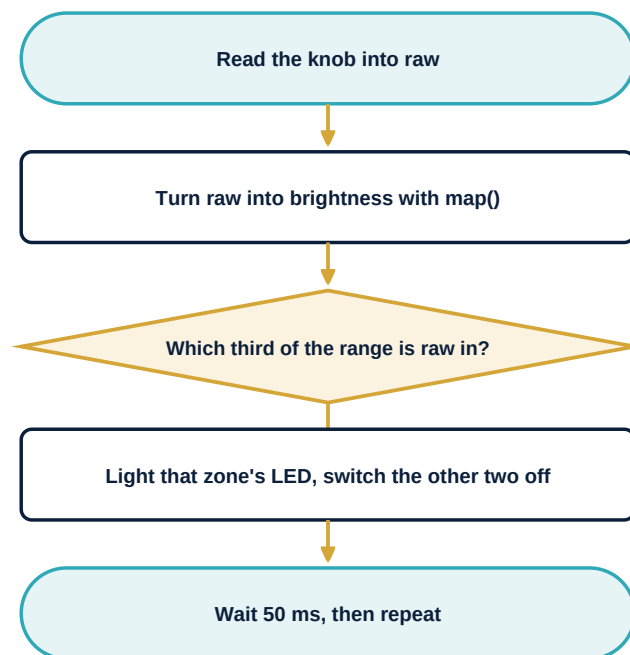


Figure 3.2 - One input choosing between three outputs

MoodLight.ino

```

int potPin = A0;
int blueLed = 9, whiteLed = 10, orangeLed = 11;

void setup() {
  pinMode(blueLed, OUTPUT);
  pinMode(whiteLed, OUTPUT);
  pinMode(orangeLed, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int raw = analogRead(potPin);
  int bright = map(raw, 0, 1023, 0, 255);

  analogWrite(blueLed, 0);      // start with all three off
  analogWrite(whiteLed, 0);
  analogWrite(orangeLed, 0);

  if (raw < 341) {
    analogWrite(blueLed, bright);    // calm zone
  } else if (raw < 682) {

```

```

    analogWrite(whiteLed, bright);    // work zone
  } else {
    analogWrite(orangeLed, bright);   // warm zone
  }

  delay(50);
}

```

Read the if - else if - else carefully. It is called a **decision ladder**, and the order matters enormously. The board checks the first condition; only if that is false does it try the second. Because of that, you never need to write *raw is between 341 and 682* - by the time the second rung is reached, the board already knows raw is at least 341.

Watch Out

Right at a boundary - around 341 or 682 - the reading wobbles by a few units and two LEDs appear to flicker against each other. That is not a fault in your code; it is a real engineering problem with a real name, and you will solve it properly in Unit 4.

Challenge

Add the buzzer to a PWM pin and use `analogWrite()` on it as well, so the knob changes light and sound together. Then try `tone(pin, frequency)` instead, with the frequency mapped from 200 to 2000, and decide which version sounds better.

New word	What it means
Potentiometer	A resistor with a slider, used as an adjustable knob.
Wiper	The middle leg of a pot, which picks up the changing voltage.
Voltage divider	A circuit that splits a voltage into a smaller share.
Decision ladder	A chain of if - else if - else, checked in order.

CHECK WHAT YOU LEARNT

Check Yourself - Unit 3**A. Choose the correct answer**

- The middle leg of a potentiometer is called the
(a) anode (b) wiper (c) trimmer (d) ground
- A pot with the knob halfway gives an analogRead of about
(a) 0 (b) 255 (c) 512 (d) 1023
- To fade an LED you must use
(a) digitalWrite (b) analogWrite (c) digitalRead (d) delay
- In an if - else if - else ladder, the conditions are checked
(a) all at once (b) in order, top to bottom (c) backwards (d) at random
- The pot's two outer legs connect to
(a) A0 and D9 (b) 5V and GND (c) GND and A0 (d) two LEDs
- Reading jumps straight from 0 to 1023. The likely fault is
(a) a burnt LED (b) the wiper on an outer leg (c) wrong baud rate (d) a missing resistor

B. Fill in the blanks

- A potentiometer works as a _____ divider.
- map(raw, 0, 1023, 0, 255) converts the range so that _____ can use it.
- Three equal zones across 0 to 1023 split at about _____ and _____.
- LEDs used for fading must be on pins marked with _____.

C. Answer in one or two lines

- Explain how turning a knob changes the voltage on the wiper.
- Why does the second rung of the ladder not need to say 'greater than 341'?
- Write down the raw values at which your own LEDs changed over.
- Name two machines in daily life that use a potentiometer.

UNIT 4**Intermediate Project: Temperature and Humidity Monitor**

Sessions 9, 10 and 11

Now a real sensor replaces your hand on the knob. The DHT11 measures two completely different quantities from one small blue box, and your machine has to decide what those numbers mean for the people sitting in the room.

By the end of this unit you will be able to:

- Install and use an external Arduino library.
- Read two different values from a single sensor.
- Build a three-band decision ladder from real measured data.
- Explain why a small gap between switching points prevents flickering.

SESSION 9

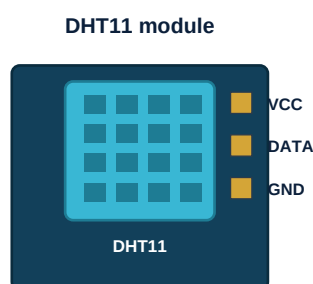
one class

The DHT11 and Your First Library

Today you will

- Learn what the DHT11 measures and how accurate it is.
- Install a library from the Library Manager.
- Print live temperature and humidity to the Serial Monitor.

Inside the blue plastic grille of a DHT11 there are two tiny sensors. One measures temperature. The other measures humidity - the amount of water vapour held in the air, given as a percentage. A small chip beside them packages both readings into a coded pulse and sends it out of a single DATA pin.



One sensor, two numbers

Temperature

how hot the air is, in degrees Celsius
useful range: 0 to 50 C

Humidity

how much water vapour the air holds, as a percentage
useful range: 20 to 90 %

The DHT11 is a digital sensor, so it goes to a digital pin - not an analog one. A library does the decoding for you.

Figure 4.1 - The DHT11 measures two different things at once

That coded pulse is far too complicated to decode by hand, and this is where a **library** comes in. A library is a bundle of instructions somebody else has already written and tested, which you are free to use. Instead of a hundred lines of timing code, you write `dht.readTemperature()` and you get a number.

- 1 In the IDE, open **Tools > Manage Libraries**.
- 2 Type **DHT sensor library** in the search box.
- 3 Choose the one by **Adafruit** and press Install.
- 4 If it offers to install **Adafruit Unified Sensor** as well, say yes. It needs it.
- 5 Close the manager. The new instructions are now available to every sketch.

ReadDHT11.ino

```
#include <DHT.h>           // bring in the library

#define DHTPIN 4           // DATA pin of the sensor
#define DHTTYPE DHT11      // which family of sensor it is
DHT dht(DHTPIN, DHTTYPE);

void setup() {
  Serial.begin(9600);
  dht.begin();
}

void loop() {
  float t = dht.readTemperature(); // degrees Celsius
  float h = dht.readHumidity();    // percentage
```

```

Serial.print("Temperature: ");
Serial.print(t);
Serial.print(" C   Humidity: ");
Serial.print(h);
Serial.println(" %");

delay(2000);          // the DHT11 needs 2 seconds between reads
}

```

Watch Out

The delay of 2000 is not optional. A DHT11 physically cannot produce a fresh reading faster than once every two seconds. Ask it more often and it hands you the old value, or the word **nan**, which stands for *not a number*.

SESSION 10

one class

Building the Monitor

Today you will

- Wire the DHT11, three LEDs and the buzzer.
- Record real classroom readings across the session.
- Decide your own cold, normal and hot boundaries.

Temperature and humidity monitor - connections

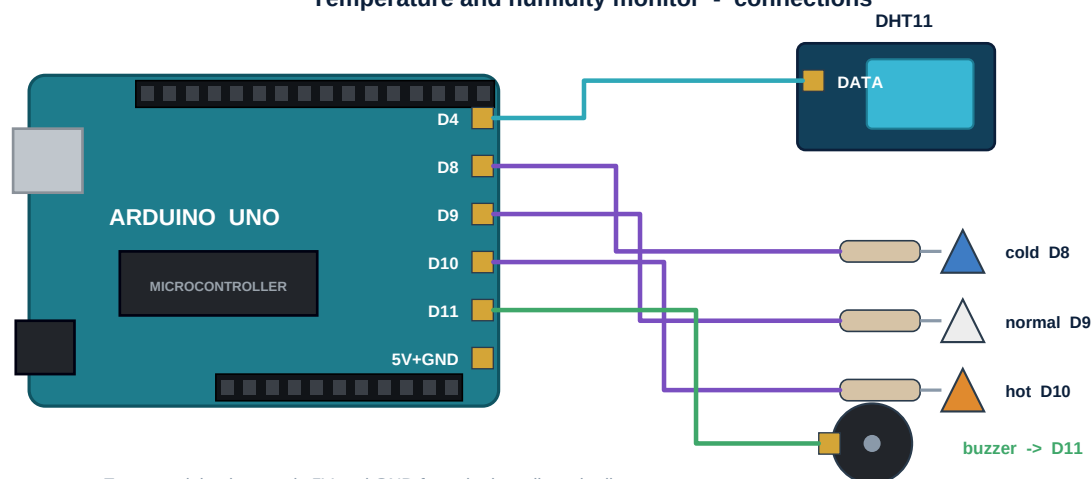


Figure 4.2 - One sensor decides which of three lamps is lit

Component pin	Arduino pin	Why
DHT11 VCC	5V	Power for the sensor
DHT11 GND	GND	Shared return path
DHT11 DATA	D4	The coded reading comes out here
Blue LED long leg	D8 through 220 ohm	Cold indicator
White LED long leg	D9 through 220 ohm	Comfortable indicator
Orange LED long leg	D10 through 220 ohm	Hot indicator
Buzzer +	D11	Alert above the hot limit
Buzzer -	GND	Return path

Before you write a single line of decision code, gather data. A machine designed on guesswork behaves like a machine designed on guesswork. Fill this table during the session.

Where and when	Temperature (C)	Humidity (%)
Classroom, start of the session	_____	_____
Classroom, end of the session	_____	_____
Near an open window	_____	_____
Sensor held in a cupped hand for 30 seconds	_____	_____
Corridor outside	_____	_____

Try This

Breathe gently on the sensor and watch the humidity climb, then fall back over the next minute. Your breath carries a great deal of water vapour. This is also a quick way to prove your sensor is genuinely working and not just repeating a stored number.

SESSION 11

one class

Alert Logic and the Decision Ladder**Today you will**

- Turn your measured data into three temperature bands.
- Add a buzzer alert above the hot limit.
- Stop the LEDs flickering at a boundary.

Now the machine gets an opinion. Using the numbers you collected, pick two boundaries: below the first is cold, above the second is hot, and everything between is comfortable. For most Indian classrooms, 24 and 30 degrees are sensible starting points - but use your own data, not ours.

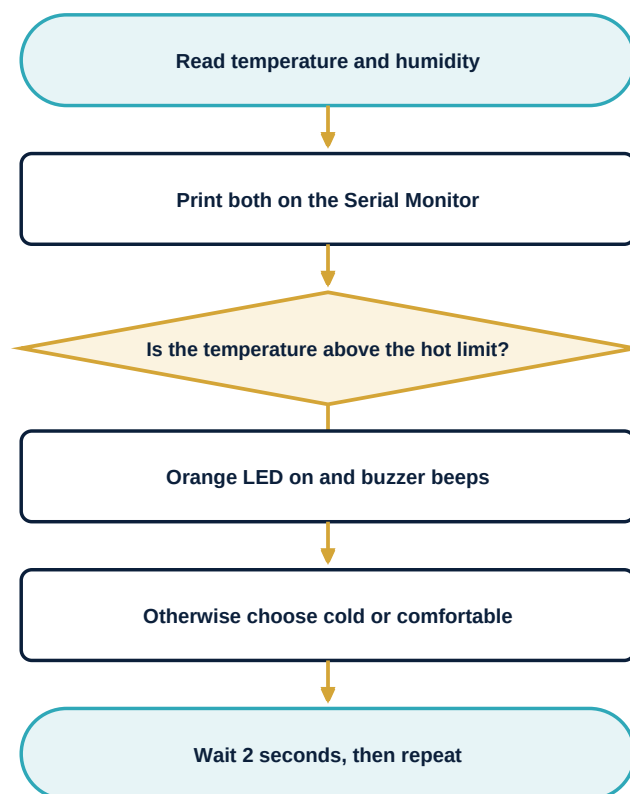


Figure 4.3 - The decision ladder, drawn before it is coded

TempHumidityMonitor.ino

```
#include <DHT.h>

#define DHTPIN 4
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);

int coldLed = 8, okLed = 9, hotLed = 10, buzzer = 11;

float coldLimit = 24.0;    // change these after your own testing
float hotLimit = 30.0;

void setup() {
  pinMode(coldLed, OUTPUT);
  pinMode(okLed, OUTPUT);
  pinMode(hotLed, OUTPUT);
  pinMode(buzzer, OUTPUT);
  Serial.begin(9600);
  dht.begin();
}

void allLedsOff() {          // our own instruction
  digitalWrite(coldLed, LOW);
  digitalWrite(okLed, LOW);
  digitalWrite(hotLed, LOW);
}

void loop() {
  float t = dht.readTemperature();
  float h = dht.readHumidity();

  if (isnan(t) || isnan(h)) {    // the sensor failed to answer
    Serial.println("Sensor not responding - check the DATA wire");
    delay(2000);
    return;                    // skip the rest of this loop
  }

  Serial.print("Temperature: ");
  Serial.print(t);
  Serial.print(" C   Humidity: ");
  Serial.print(h);
  Serial.println(" %");

  allLedsOff();

  if (t < coldLimit) {
    digitalWrite(coldLed, HIGH);
    digitalWrite(buzzer, LOW);
  } else if (t < hotLimit) {
    digitalWrite(okLed, HIGH);
    digitalWrite(buzzer, LOW);
  } else {
    digitalWrite(hotLed, HIGH);
    digitalWrite(buzzer, HIGH);    // too hot - warn the room
    delay(300);
    digitalWrite(buzzer, LOW);
  }

  delay(2000);
}
```

Two professional touches are hiding in that program. The first is `isnan(t)`, which checks whether the sensor actually answered. Real engineers never assume a sensor replied. The second is `allLedsOff()`, an instruction we wrote ourselves so that the ladder below stays short and readable.

11.1 The Flickering Problem, and Its Real Name

Sit the sensor at exactly 30.0 degrees and watch. The reading wobbles - 29.9, 30.1, 30.0, 30.2 - and the orange LED and the buzzer flicker on and off maddeningly. Real thermostats solve this with a trick called **hysteresis**: the switch-on point and the switch-off point are deliberately made different.

In plain English: once the alarm has started, do not stop it until the temperature has fallen a clear degree below the limit. A one-degree gap is enough to make the machine feel calm and deliberate instead of nervous.

Challenge

Implement hysteresis. Add `boolean alarmOn = false;` at the top. Switch the alarm on when `t > hotLimit`, but switch it off only when `t < hotLimit - 1.0`. Watch the flickering disappear completely.

New word	What it means
Library	Ready-made instructions written by somebody else that you can use.
Humidity	How much water vapour the air is holding, as a percentage.
nan	Not a number - what a sensor returns when it fails to answer.
Hysteresis	Using different switch-on and switch-off points to stop flickering.

CHECK WHAT YOU LEARNT

Check Yourself - Unit 4**A. Choose the correct answer**

- The DHT11 measures
(a) only temperature (b) temperature and humidity (c) only humidity (d) light and heat
- A DHT11 can give a fresh reading roughly once every
(a) 2 milliseconds (b) 2 seconds (c) 2 minutes (d) 2 hours
- A library is
(a) a type of sensor (b) ready-made code you can use (c) a wiring diagram (d) an error
- isnan(t) checks whether
(a) t is too hot (b) the sensor failed to answer (c) t is negative (d) the LED is on
- Humidity is measured in
(a) degrees (b) percent (c) ohms (d) volts
- Using different switch-on and switch-off points is called
(a) calibration (b) hysteresis (c) mapping (d) debugging

B. Fill in the blanks

- The DHT11 sends its reading out of the _____ pin.
- A temperature value should be stored in a variable of type _____.
- Adafruit's DHT library is installed from Tools > Manage _____.
- A one-degree gap between switching points stops the LED from _____.

C. Answer in one or two lines

- Write the cold and hot limits you chose, and the data that led you to them.
- Why is it a mistake to ask a DHT11 for a reading every 100 milliseconds?
- Explain hysteresis to a Class 6 student in two sentences.
- Why did we write allLedsOff() instead of repeating three lines each time?

UNIT 5

Final Project: Smart Rain Alarm System

Sessions 12, 13, 14 and 15

Your final machine uses two senses at once. It detects not just whether it is raining, but how heavily - and it stays sensibly quiet at night. This is the project you will demonstrate.

By the end of this unit you will be able to:

- Explain how a rain drop sensor turns water into a number.
- Combine two analog sensors in one decision.
- Design alert patterns that tell severity by sound alone.
- Calibrate a machine using real water and defend your numbers.

SESSION 12

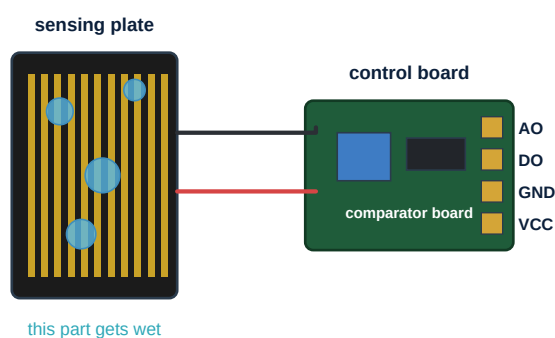
one class

How a Rain Sensor Works

Today you will

- Understand the sensing plate and the control board.
- Learn why AO is more useful than DO for this project.
- Agree the team roles and plan the build.

The black plate is covered with two gold tracks that weave between each other but never touch. Dry, no electricity can cross the gap. When water lands on the plate it bridges those tracks, because rainwater conducts electricity slightly. A little water makes a poor bridge; a lot of water makes a good one. That difference is exactly what lets your machine judge how heavy the rain is.

**Two boards, one sensor****AO - analog out**

a number from 0 to 1023: how wet the plate is

DO - digital out

just wet or dry, set by the small blue screw

In this project you use AO, because you want to know how heavy the rain is, not merely whether it is raining.

Figure 5.1 - The rain sensor comes as two pieces joined by two wires

Notice the choice you are making. The DO pin would tell you only *raining* or *not raining* - a Class 6 answer. The AO pin gives you the full range, so your alarm can distinguish a drizzle from a downpour. That is a Class 7 answer, and it is the whole reason this project is worth building.

Watch Out

Only the plate may get wet. Keep the green control board, the breadboard and the Arduino well away from the water, and mount the plate at a slight angle so drops run off instead of pooling. A pool of water reads as heavy rain long after the rain has stopped.

Role	Responsible for	Name
Builder	Mounting the plate at an angle and keeping water away from the electronics	_____
Wiring engineer	Every connection, checked twice before power on	_____
Programmer	Typing, uploading and adjusting the threshold values	_____
Recorder	The notebook: readings, faults, and what fixed them	_____

SESSION 13

one class

Building the Alarm**Today you will**

- Wire the rain sensor, the LDR, three LEDs and the buzzer.
- Understand why the LDR is included at all.
- Get a first working version running.

Smart rain alarm - connections

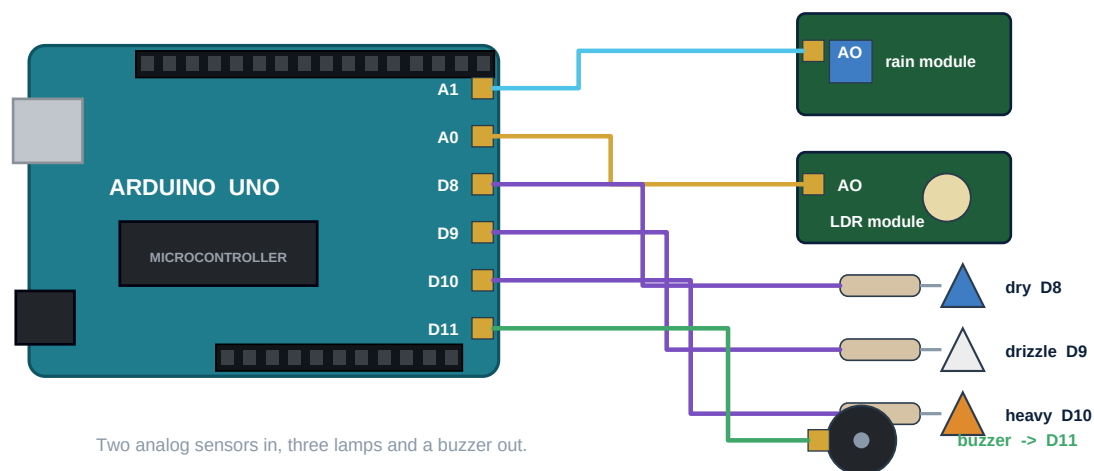


Figure 5.2 - Two senses working together

Component pin	Arduino pin	Why
Rain module AO	A1	How wet the plate is, 0 to 1023
Rain module VCC / GND	5V / GND	Power
LDR module AO	A0	How bright it is outside
LDR module VCC / GND	5V / GND	Power
Blue LED long leg	D8 through 220 ohm	Dry
White LED long leg	D9 through 220 ohm	Drizzle
Orange LED long leg	D10 through 220 ohm	Heavy rain
Buzzer +	D11	The alarm itself

Why the LDR? Because a good machine is considerate. An alarm that shrieks at three in the morning gets unplugged, and an unplugged alarm protects nobody. The LDR lets your system show the lamps silently at night while still sounding the buzzer in daylight, when somebody might actually act on the warning.

Did You Know

This idea has a name in engineering: **context awareness**. A phone that stays silent during a calendar meeting is doing exactly what your rain alarm does - using a second piece of information to decide how loudly to behave.

SESSION 14

one class

Calibration and Alarm Patterns**Today you will**

- Measure real dry, drizzle and heavy readings with water.
- Choose and justify your two thresholds.
- Give each level its own buzzer pattern.

Calibration means tuning a machine to the world it actually lives in. Your plate, your water and your room are not ours, so our numbers are only a starting point. Use a dropper or a wet finger and fill this table honestly before you set any threshold.

Condition of the plate	Reading on A1	What it should mean
Completely dry	_____	DRY
Three or four drops	_____	DRIZZLE
Sprayed all over	_____	HEAVY
Wiped dry again	_____	back to DRY
drizzleLevel you chose	_____	boundary 1
heavyLevel you chose	_____	boundary 2

Watch Out

Some rain modules count **up** as they get wetter and others count **down**. Find out which yours does before you write a single if statement. If your alarm behaves backwards, do not rewire anything - swap the > signs for <.

SmartRainAlarm.ino

```
int rainPin = A1;
int ldrPin  = A0;
int dryLed = 8, drizzleLed = 9, heavyLed = 10;
int buzzer = 11;

int drizzleLevel = 400;  // change these after your own testing
int heavyLevel   = 700;
int nightLevel   = 300;  // below this, treat it as night

void setup() {
  pinMode(dryLed, OUTPUT);
  pinMode(drizzleLed, OUTPUT);
  pinMode(heavyLed, OUTPUT);
  pinMode(buzzer, OUTPUT);
  Serial.begin(9600);
}

void allLedsOff() {
  digitalWrite(dryLed, LOW);
  digitalWrite(drizzleLed, LOW);
  digitalWrite(heavyLed, LOW);
}

void beep(int times, int gap) {  // our own alarm pattern
  for (int i = 0; i < times; i++) {
    digitalWrite(buzzer, HIGH);
    delay(120);
    digitalWrite(buzzer, LOW);
    delay(gap);
  }
}

void loop() {
  int wet  = analogRead(rainPin);
  int light = analogRead(ldrPin);
  boolean daytime = (light > nightLevel);

  Serial.print("wet: ");
  Serial.print(wet);
  Serial.print("  light: ");
  Serial.println(light);

  allLedsOff();

  if (wet < drizzleLevel) {
```

```

    digitalWrite(dryLed, HIGH);           // no rain, stay silent
  } else if (wet < heavyLevel) {
    digitalWrite(drizzleLed, HIGH);
    if (daytime) { beep(1, 800); }       // one slow beep
  } else {
    digitalWrite(heavyLed, HIGH);
    if (daytime) { beep(4, 120); }       // fast urgent beeps
  }

  delay(400);
}

```

The instruction `beep(times, gap)` is one you wrote yourself, and it takes two pieces of information: how many beeps, and how long to pause between them. That is what makes drizzle and downpour sound genuinely different. Somebody in the next room can tell how hard it is raining without looking at your machine at all - and that is good design.

SESSION 15

one class

Testing and Your Demonstration

Today you will

- Find and fix the faults in your own machine.
- Pass every acceptance test with real water.
- Present your project in three minutes.

Problem you may see	Most likely cause	What to do
Alarm stays on after the plate is dried	Water pooling between the tracks	Mount the plate at a steeper angle and wipe it properly
Readings jump about wildly	A loose jumper on A1, or a wet control board	Reseat the wire; keep the green board dry
Buzzer never sounds	The LDR thinks it is night	Print light and check it against nightLevel
Alarm behaves backwards	Your module counts down as it gets wetter	Swap the > signs for < in the ladder
Nothing works at all	Missing shared GND	One GND wire must reach every module

Acceptance test	Result	Tick
Dry plate: blue LED on, buzzer silent	_____	■
A few drops: white LED on, one slow beep	_____	■
Sprayed plate: orange LED on, fast beeps	_____	■
Plate wiped dry: returns to blue within 5 seconds	_____	■
Sensor covered to simulate night: lamps still work, buzzer silent	_____	■

15.1 Your Three-Minute Demonstration

- The problem** - one sentence on what your machine solves.
- How it senses** - hold up the plate and explain how water bridges the gold tracks.
- How it decides** - show the two threshold numbers and say where they came from.
- How it acts** - run it live with a water spray. Let the beeps tell the story.
- The clever bit** - explain why the LDR is there. This is what makes your project stand out.

6 What went wrong - name one fault you found and fixed.

Challenge

Add the SG90 servo as a model window shutter or clothesline cover. Wire its orange signal wire to D6, include the Servo library, and use `shutter.write(90);` to close it when the reading crosses `heavyLevel` and `shutter.write(0);` to reopen it once the plate is dry.

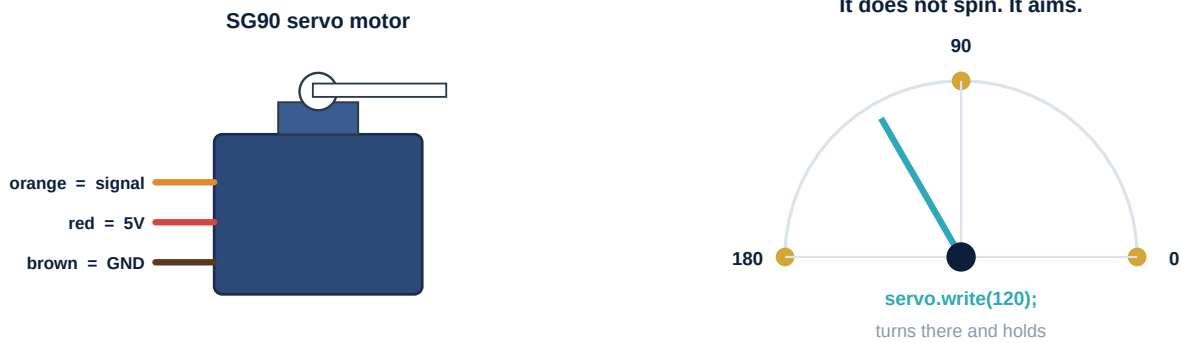


Figure 5.3 - A servo turns to an exact angle between 0 and 180 degrees

New word	What it means
Conductivity	How easily electricity passes through something. Water conducts a little.
Threshold	The number at which a decision changes from one answer to the other.
Calibration	Adjusting a machine to suit the real place it works in.
Context awareness	Using a second piece of information to decide how to behave.

CHECK WHAT YOU LEARNT

Check Yourself - Unit 5**A. Choose the correct answer**

- The rain sensor works because water
(a) is cold (b) conducts electricity (c) is heavy (d) reflects light
- For judging how heavy the rain is, you should use
(a) the DO pin (b) the AO pin (c) the VCC pin (d) the GND pin
- The LDR is included in this project so that
(a) the plate dries faster (b) the alarm stays quiet at night (c) the LEDs are brighter (d) the code is shorter
- beep(4, 120) means
(a) beep for 4 seconds (b) four beeps with short gaps (c) beep at 120 volts (d) wait 4 minutes
- If your alarm behaves backwards you should
(a) rewire everything (b) swap the > signs for < (c) buy a new sensor (d) add a resistor
- Water pooling on the plate makes the machine report
(a) dry for too long (b) heavy rain for too long (c) nothing at all (d) night time

B. Fill in the blanks

- Only the _____ of the rain sensor is allowed to get wet.
- Two analog sensors in this project are connected to pins _____ and _____.
- Tuning a machine to its real surroundings is called _____.
- Using a second piece of information to decide how to behave is called _____ awareness.

C. Answer in one or two lines

- Write the three readings you measured - dry, drizzle and sprayed - and the two thresholds you chose between them.
- Explain in two sentences how water lets electricity cross the plate.
- Why is an alarm that wakes people at 3 am a badly designed alarm?
- Which of the three projects this year taught you the most, and why?

Extra Challenges

Finished early? Pick any of these. They all use parts you already have.

#	Challenge	Uses
1	A knob-controlled buzzer whose pitch rises as you turn it	pot + tone()
2	A rainbow fader that cycles brightness up and down on its own	for loop + PWM
3	A comfort meter that combines temperature and humidity into one score	DHT11 + maths
4	A thermostat with hysteresis that never flickers at the boundary	DHT11 + boolean
5	A humidity alarm for a book cupboard, silent below 60 percent	DHT11 + buzzer
6	A night lamp that also reports the light level every ten seconds	LDR + Serial
7	A rain gauge that counts how many minutes it rained today	rain module + millis()
8	An automatic window shutter driven by the servo	rain module + servo
9	A two-sensor weather station printing a neat table to Serial	DHT11 + LDR
10	A knob that sets the alarm threshold, so no re-uploading is needed	pot + rain module

MIXED REVISION

Twenty Questions to Test Yourself

- 1 What is the difference between a digital and an analog signal?
- 2 What value has a resistor banded red - red - brown?
- 3 Which pin of a module always goes to 5V?
- 4 What is the largest number `analogRead()` can return?
- 5 What is the largest number `analogWrite()` will accept?
- 6 Write the `map()` line that converts 0 to 1023 into 0 to 255.
- 7 Name three PWM-capable pins on the Arduino Uno.
- 8 Explain in one sentence how PWM dims an LED.
- 9 Which leg of a potentiometer goes to the analog pin?
- 10 What does `Serial.begin(9600)` do?
- 11 Why must the Serial Monitor's baud number match your code?
- 12 Which two quantities does a DHT11 measure?
- 13 Why must you wait two seconds between DHT11 readings?
- 14 What does `isnan()` check for?
- 15 What is a library, and why is it useful?
- 16 What is hysteresis, and what problem does it solve?
- 17 How does water let electricity cross the rain sensor plate?
- 18 Why is AO more useful than DO in the rain alarm?
- 19 Why does the rain alarm include a light sensor?
- 20 Name one thing you would add to your rain alarm with one more session.

Words to Know

Word	Meaning
Analog	A value that can be anything within a range, not just on or off.
analogRead()	Reads an analog pin and returns a number from 0 to 1023.
analogWrite()	Sends a PWM output, accepting a number from 0 to 255.
AO	The analog-out pin of a module. Gives a full reading.
Baud rate	The speed at which the board and the laptop talk. Usually 9600.
Calibration	Adjusting a machine to suit the real place it works in.
Conductivity	How easily electricity passes through a material.
Context awareness	Using extra information to decide how a machine should behave.
Decision ladder	A chain of if - else if - else, checked in order from the top.
Digital	A signal with only two values: HIGH or LOW.
DO	The digital-out pin of a module. Gives only yes or no.
float	A variable that holds a number with a decimal point.
Humidity	How much water vapour the air holds, given as a percentage.
Hysteresis	Different switch-on and switch-off points, used to stop flickering.
int	A variable that holds a whole number.
Library	Ready-made code written by someone else that you can include and use.
map()	Stretches one range of numbers neatly onto another.
Module	A small ready-made board with components already fitted.
nan	Not a number. What a sensor returns when it fails to answer.
Ohm	The unit of resistance.
Potentiometer	A resistor with a slider, used as an adjustable knob.
PWM	Pulse Width Modulation. Fast switching that makes an output look in-between.
Serial Monitor	The window where the board prints what it is thinking.
Servo	A motor that turns to an exact angle and holds it.
Threshold	The number at which a decision changes from one answer to the other.
Trimmer	A small adjustable screw that sets a module's DO threshold.
VCC	The power-in pin of a module. Connects to 5V.
Voltage divider	A circuit that splits a voltage into a smaller share.
Wiper	The middle leg of a potentiometer.

Answer Key

Section A only. Sections B and C are for you and your trainer to discuss - there is often more than one good answer.

Check Yourself - Unit 1

Q	Answer	The correct option
1	(b)	digital
2	(b)	analog
3	(b)	220 ohm
4	(b)	5V
5	(b)	a number from 0 to 1023
6	(b)	the DO threshold

Check Yourself - Unit 2

Q	Answer	The correct option
1	(b)	0 and 1023
2	(b)	0 and 255
3	(b)	convert one range to another
4	(b)	switching on and off very fast
5	(c)	D9
6	(b)	setup()

Check Yourself - Unit 3

Q	Answer	The correct option
1	(b)	wiper
2	(c)	512
3	(b)	analogWrite
4	(b)	in order, top to bottom
5	(b)	5V and GND
6	(b)	the wiper on an outer leg

Check Yourself - Unit 4

Q	Answer	The correct option
1	(b)	temperature and humidity
2	(b)	2 seconds
3	(b)	ready-made code you can use
4	(b)	the sensor failed to answer
5	(b)	percent
6	(b)	hysteresis

Check Yourself - Unit 5

Q	Answer	The correct option
1	(b)	conducts electricity
2	(b)	the AO pin
3	(b)	the alarm stays quiet at night
4	(b)	four beeps with short gaps
5	(b)	swap the > signs for <
6	(b)	heavy rain for too long

My Project Log

Fill one row after every class. This is your evidence, and it counts towards your assessment.

Session	What I built or learnt	A problem I hit	How I fixed it
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			

A last word

This year your machines stopped answering yes or no and started answering how much. That is a bigger step than it sounds. Everything from a thermostat to a self-driving car is built on exactly the idea you used in these three projects: read a number, decide what it means, and act sensibly.